

Neo Cloud Room (CloudKitchen Platform) - Comprehensive System Analysis & Workflow Architecture Document

1. Executive Summary & Project Architecture

Neo Cloud Room (also known as *CloudKitchen*) is a multi-role, multi-vendor, enterprise-grade digital platform designed to handle two primary service domains:

- Food & Culinary Services:** Cloud Kitchens, Mess Providers, and Bakeries delivering fresh meals to local customers based on pincodes and real-time menu availability.
- Property & Room Rental Services:** Short-term and long-term room listings, capacity management, booking requests, date-conflict validation, and payment settlement.
- Furniture Exploration & Direct Store Inquiries:** Showcasing catalog items with query routing.

Core Technology Stack

- Frontend:** Next.js 15+ (App Router), React 19, TypeScript, Tailwind CSS, Lucide Icons, Modern Responsive UI.
 - Backend API:** Next.js API Routes (`/api/ ...`), Node.js, TypeScript, NextAuth.js / JWT Auth.
 - Database & ORM:** PostgreSQL database managed via Prisma ORM (`prisma/schema.prisma`).
 - Payment Processing:** Razorpay Integration (HMAC-SHA256 signature verification for Online payments, COD workflow with rider cash collection).
 - File Storage:** Local & Cloud multipart form upload handler (`lib/upload.ts`) for document verification (Adhaar, FSSAI, Light Bill, Passbook) and media assets (Kitchen, Cuisine, Room photos, Banners).
 - Location & Routing:** Geocoding engine and pincode-matching delivery verification.
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2. Master System Enums & Data States

Domain	Enum / Field Name	Allowed Values / States	Description
System Roles	User.role	SUPERADMIN , AGENT , SUPPORT , SELLER , USER , DELIVERY	Defines access control permissions and available dashboards.
Seller Business Category	SellerProfile.businessCategory	FOOD , PROPERTY , BOTH	Determines whether the store operates as a Cloud Kitchen/Mess, Room Host, or Both.
Seller Food Type	SellerProfile.foodType	VEG , BOTH	Defines dietary compliance for food offerings.
Seller Verification	verificationStatus foodVerificationStatus propertyVerificationStatus	PENDING , APPROVED , REJECTED , REVISION , NONE	Lifecycle state of vendor document verification and audit resubmission.
Order Status	Order.status	PENDING , PREPARING , OUT_FOR_DELIVERY , DELIVERED , CANCELLED	Order progression state from placement to delivery.
Payment Method	paymentMethod	ONLINE , COD , QR	Payment channel chosen by customer or rider.
Booking Status	Booking.status	PENDING , CONFIRMED , CANCELLED	Progression state of property room bookings.
Subscription Status	Subscription.status	ACTIVE , EXPIRED , INACTIVE	Vendor subscription state allowing store operational status.
Coupon Status	approvalStatus	PENDING_APPROVAL , APPROVED , REJECTED , REVISION	Approval pipeline for seller-created discount coupons and popup banners.
Support Ticket Status	Ticket.status	OPEN , IN_PROGRESS , RESOLVED , CLOSED	Progression of customer support tickets.
Refund Request	Refund.status	PENDING , APPROVED , REJECTED	Status of money refund

Domain	Enum / Field Name	Allowed Values / States	Description
Status			resubmissions.
Delivery Transactions	DeliveryTransaction.type	COD_COLLECTION , SETTLEMENT , ADJUSTMENT	Delivery wallet ledger types tracking cash movement between driver and seller.

3. Detailed Role-Based Features & Flow Breakdown



3.1 SUPERADMIN (Master Platform Administrator)

- **Global Governance:** Complete access to all backend resources and administrative dashboard controls.
- **Verification & Approval Engine:**
 - Reviews and audits initial seller registrations, document attachments (Adhaar, FSSAI, Light Bill, Passbook), resubmissions, and category application expansions.
 - Approves, rejects, or flags resubmissions with administrative notes (`adminNote`).
 - Audits pending seller-requested coupons and pop-up promotional banners (`PENDING_APPROVAL` → `APPROVED` / `REJECTED`).
- **Category & Taxonomy Management:**
 - Creates and manages top-level platform categories (`FOOD` vs `ROOM`).
 - Creates, edits, and organizes `FoodCategory` and `FoodSubCategory` with image upload handling.
- **Subscription Plan & Coupon Creation:**
 - Configures vendor subscription plans (`SubscriptionPlan`) with parameters: `name` , `price` , `durationMonths` , `category` (`FOOD` , `PROPERTY` , `BOTH`), and JSON feature lists.
 - Creates global or plan-specific `SubscriptionCoupon` codes with percentage or flat discounts, expiry dates, and maximum usage caps.
- **Financial & Refund Management:**
 - Inspects refund resubmissions (`Refund`) for paid orders and room bookings.
 - Approves refunds (triggering status update `isPaid = false` on orders or `status = CANCELLED` on bookings) or rejects requests with transaction references.
 - Monitors system-wide audit logs (`AuditLog`) capturing administrative actions.
- **User, Agent & Support Personnel Governance:**
 - Creates and manages field Agents (`AGENT`) and Customer Support Staff (`SUPPORT`) accounts.
 - Promotes or demotes user accounts, inspects user activity logs, and toggles user active status.

3.2 AGENT / ADMIN (Field / Branch Administrator)

- **Assigned Seller Management:**
 - Directly manages a subset of assigned seller profiles (`assignedSellers`).
 - Verifies submitted identity cards and regulatory proofs for newly onboarding vendors in their region.
 - **Promotions & Banner Operations:**
 - Creates and updates store-level and global pop-up banners.
 - If an Agent creates a global banner, it is queued as `PENDING_APPROVAL` for Superadmin authorization; store-level banners created by agents are auto-approved (`APPROVED`).
 - **Fleet Wallet Adjustments:**
 - Adjusts delivery driver outstanding COD balances (`INCREMENT` or `DECREMENT`) with detailed reason logs (`ADJUSTMENT` transactions).
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3.3 SUPPORT STAFF (Customer Support & Refund Operations Administrator)

- **Dedicated Administrative Role:**
 - Created by Superadmin (`role = SUPPORT`) to manage customer support operations, user dispute resolution, and payment refunds via `/auth/login/admin` .
- **Support Ticket Resolution & Live Messaging:**
 - Access to dedicated support ticket dashboard (`/dashboard/support/tickets`).
 - Queries all system-wide support tickets across customers and vendors (`listTickets`).
 - Real-time threaded ticket messaging with users (`sendTicketMessage`), appending staff responses.
 - Ticket lifecycle control: Updates ticket status (`OPEN` → `IN_PROGRESS` → `RESOLVED` / `CLOSED`).
 - Ability to create support tickets on behalf of users when handling call/chat inquiries.
- **Refund Request Audit & Money Fulfillment:**
 - Access to refund processing portal (`/dashboard/support/refunds`).
 - Audits pending refund requests (`Refund`) for paid orders and room bookings across the platform.
 - Authorizes refunds (`APPROVED`) or rejects invalid claims (`REJECTED`) with bank transaction references (`transactionId`) and internal administrative notes (`adminNote`).
 - Executing a refund approval automatically updates order payment state (`isPaid = false`) or cancels room bookings (`status = CANCELLED`).
- **Order Inspection & Cancellation Override:**
 - Full access to inspect details of any order (`getOrderDetails`).
 - Administrative authority to cancel problematic or disputed orders (`cancelOrder`), bypassing customer self-cancel constraints and automatically queueing refund requests if the order was prepaid.
- **User Activity Audit:**
 - Inspects registered user list (`/api/superadmin/users`) and monitors individual user activity logs (`/api/superadmin/users/[id]/activity`).
- **Category & Taxonomy Read/Write Access:**

- Views and updates food categories and subcategories to assist vendors with menu categorization disputes.

3.4 SELLER / VENDOR (Cloud Kitchen, Mess, Bakery, Room Host)

- **Multi-Step Onboarding & Document Verification Flow:**
 - Registers business profile with business name, seller type (Homely Food, Mess, Bakery), location details (flat, locality, landmark, city, pincode), food classification (`VEG` / `BOTH`), and business category (`FOOD` , `PROPERTY` , `BOTH`).
 - Uploads required legal proofs: Adhaar Card (front & back), FSSAI License, Electricity/Light Bill, Bank Passbook, along with Kitchen, Cuisine, and Room photos.
 - Receives a unique store tracking ID (e.g., `SHOP-X7K2P9`).
- **Reapply / Revision Resubmission Workflow:**
 - If Superadmin/Agent flags profile status as `REVISION` with feedback notes (`verificationNote`), seller dashboard enters Revision Mode.
 - Seller resubmits updated/corrected documents and resends verification request, which resets status to `PENDING` for re-auditing.
- **Category Expansion Application Flow:**
 - Existing approved food sellers can apply to add `PROPERTY` offerings (and vice versa) via `/api/seller/category-application` .
 - Seller submits category-specific proof files (e.g., room photos for property expansion, FSSAI for food expansion). Verification status updates to `PENDING` for that category.
- **Subscription Management & Activation:**
 - Stores require an active `Subscription` matching their operational domain (`FOOD` or `PROPERTY`) to open for orders/bookings.
 - Seller selects plan, applies discount coupons (`SubscriptionCoupon`), and completes Razorpay payment.
 - Stacking logic: Renewing an existing active plan automatically extends the `validUntil` expiry date from the current expiration date.
- **Store & Inventory Controls:**
 - Toggles store Online/Offline state (`isOnline`).
 - Sets day-wise detailed operational hours (`operationalHours` JSON schema containing `isOpen` , `openTime` , `closeTime` for Mon-Sun).
 - Manages served pincodes (`ServedPincode`).
 - Adds/edits food items: Name, description, price, food type (`VEG` / `NON_VEG`), categories, stock quantity (-1 for infinite, or numeric counter with auto-decrement), and item-specific delivery pincodes.
 - Adds/edits room listings: Title, capacity, night price, description, images, and availability status.
- **Order Fulfillment & Delivery Fleet Operations:**
 - Real-time order dashboard tracking statuses: `PENDING` → `PREPARING` → `OUT_FOR_DELIVERY` → `DELIVERED` .
 - Creates dedicated delivery personnel accounts (`DeliveryPerson`) with 10-digit phone verification and login credentials.

- Assigns delivery riders to specific orders (automatically updates order state to `OUT_FOR_DELIVERY`).
 - Manages COD cash collections: Accepts cash collected by delivery riders, records `SETTLEMENT` transactions, and decrements rider's `outstandingBalance` .
 - **Promotions & Reviews:**
 - Requests custom discount coupons and store pop-up banners (submitted for Admin/Superadmin approval).
 - Views order reviews and item-level customer ratings.
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3.5 USER / CUSTOMER

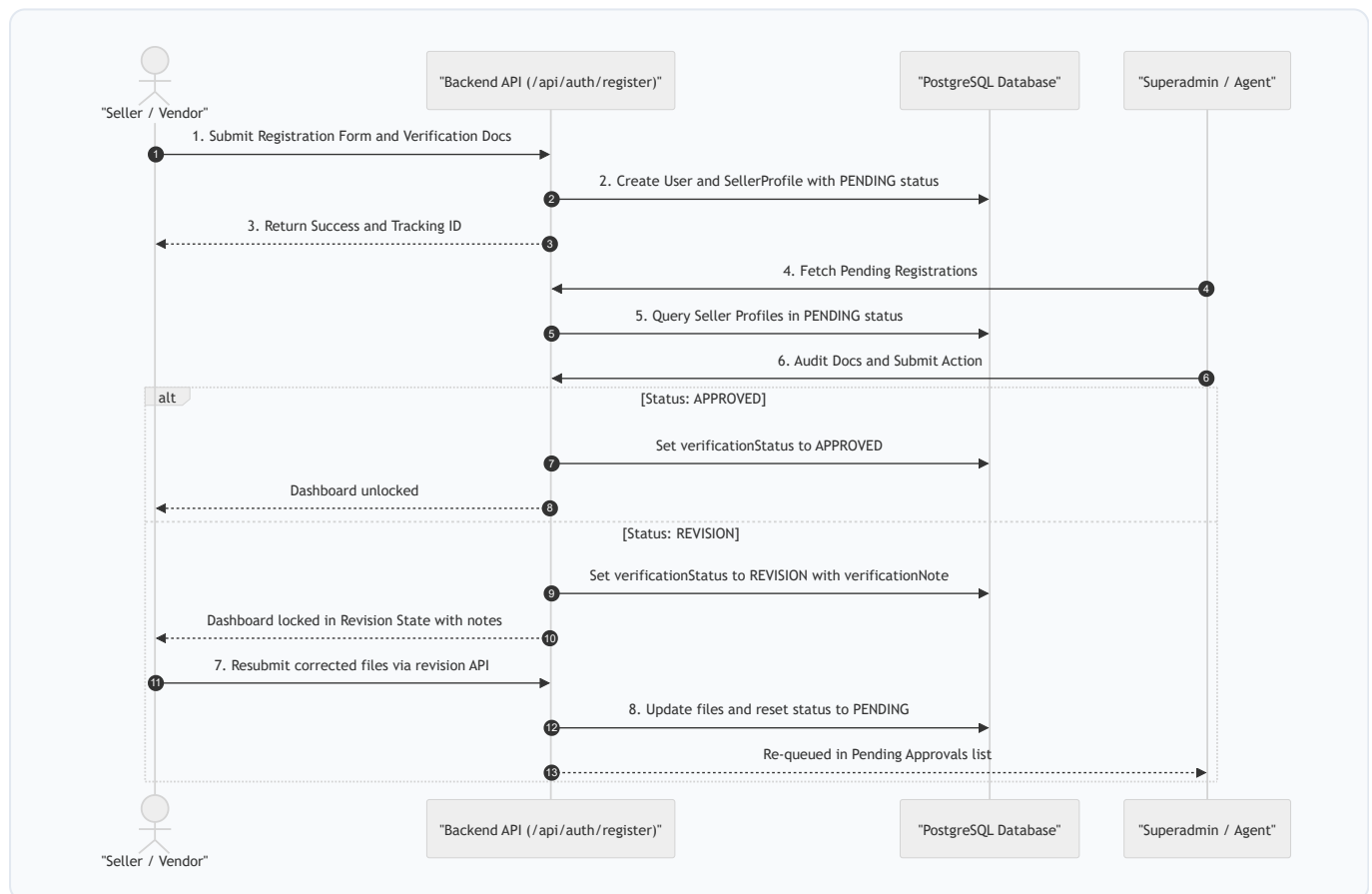
- **Profile & Saved Address Management:**
 - Creates customer account, manages contact information.
 - Saves multiple delivery addresses (`Address`) labeled Home, Work, or Other with pincodes and latitude/longitude geocoding. Sets default delivery address.
 - **Multi-Service Catalog Exploration:**
 - Explores nearby food items, mess plans, bakeries, and rooms based on selected delivery pincode.
 - Live system validations prevent selecting offline stores, non-delivering pincodes, or items outside operational store hours.
 - Explores available furniture catalog and submits direct inquiries via `/api/furniture/query` .
 - **Checkout, Coupon Redemption & Order Placement:**
 - Cart engine validates stock availability, seller subscription validity, store open hours, and pincode matching.
 - Validates coupons against rule engines: minimum cart value (`minimumCartValue`), max usage per user (`maxUsagesPerUser`), global max users (`maxUsers`), and store category compatibility.
 - Selects payment mode: Online Payment (Razorpay checkout) or Cash on Delivery (COD).
 - **Real-Time Order & Booking Tracking:**
 - Tracks order progress (`PENDING` , `PREPARING` , `OUT_FOR_DELIVERY` , `DELIVERED`).
 - Views printable tax invoices (`/invoice/order/[id]`).
 - **Property Room Booking Flow:**
 - Selects check-in (`startDate`) and check-out (`endDate`) dates.
 - System checks for overlapping confirmed/pending bookings for the chosen room.
 - Submits booking request (`PENDING`). Once host confirms, customer completes Online or COD payment.
 - **Support, Tickets & Refunds:**
 - Files support tickets (`Ticket`) categorized under `FOOD` , `ROOM` , `PAYMENT` , or `OTHER` .
 - Participates in live threaded ticket chat with support staff.
 - Requests order or booking cancellations (if order is `PENDING`) and submits refund claims for paid cancelled orders.
 - **Ratings & Reviews:**
 - Submits overall 1 to 5-star order reviews with optional text comments.
 - Submits granular item-level star ratings (`ItemRating`) for each food item included in the delivered order.
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3.6 DELIVERY PERSON (Rider Fleet)

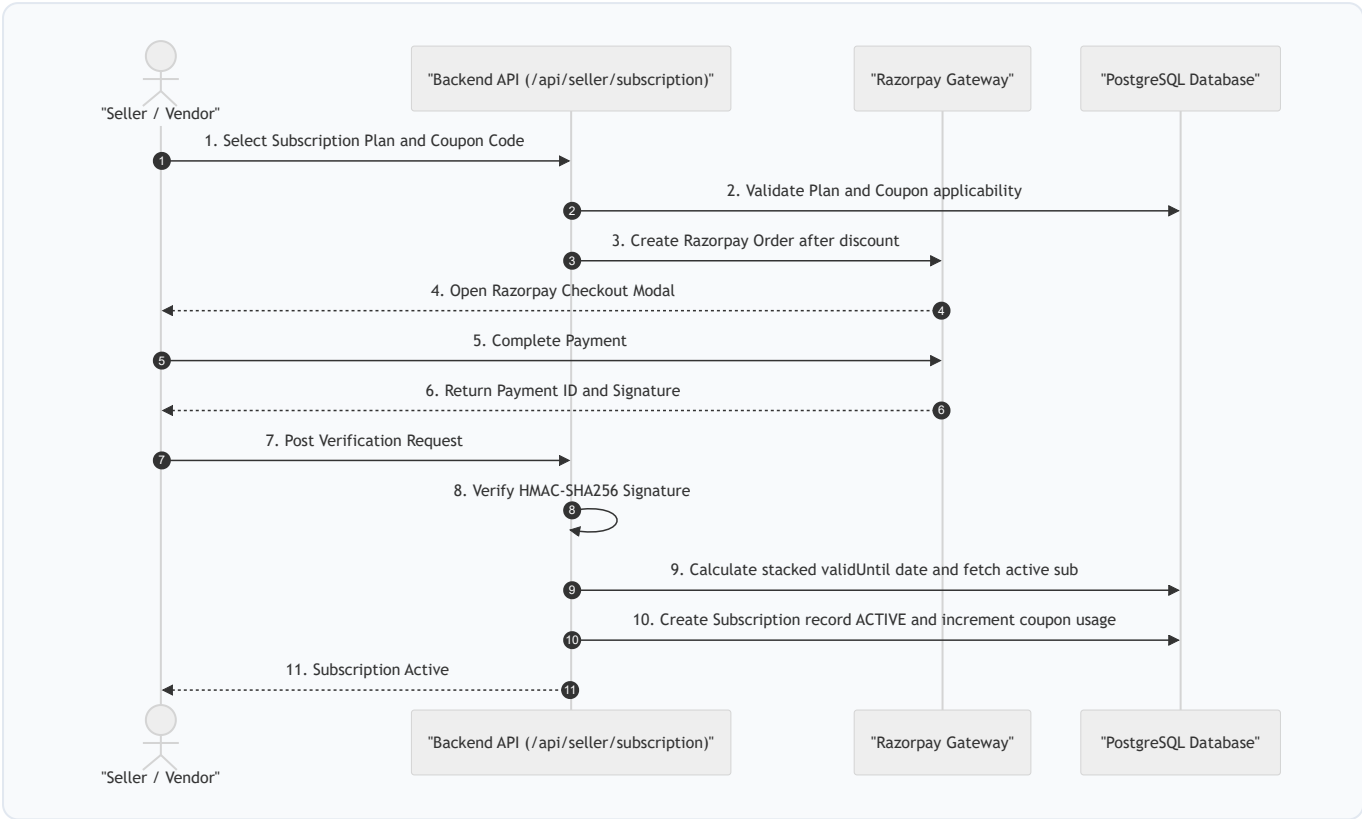
- **Account Access:** Logins using credentials created by their employing Seller.
- **Delivery Management Dashboard:**
 - Views orders assigned to them by the seller (`deliveryPersonId`).
 - Updates order status: `OUT_FOR_DELIVERY` → `DELIVERED` .
- **Payment & Wallet Mechanics:**
 - For Cash on Delivery (COD) orders: Upon marking an order as `DELIVERED` , the system automatically creates a `COD_COLLECTION` transaction and increments the rider's `outstandingBalance` by the order total amount.
 - For Doorstep Online Payments: Can generate a doorstep Razorpay payment QR code for the customer to scan and pay. Upon successful verification, order is marked `isPaid = true` and `DELIVERED` .
- **Cash Settlement:** Handovers collected cash to the seller, who logs a `SETTLEMENT` action to clear the driver's outstanding wallet balance.

4. End-to-End Core Workflows (Sequence Diagrams & Step-by-Step)

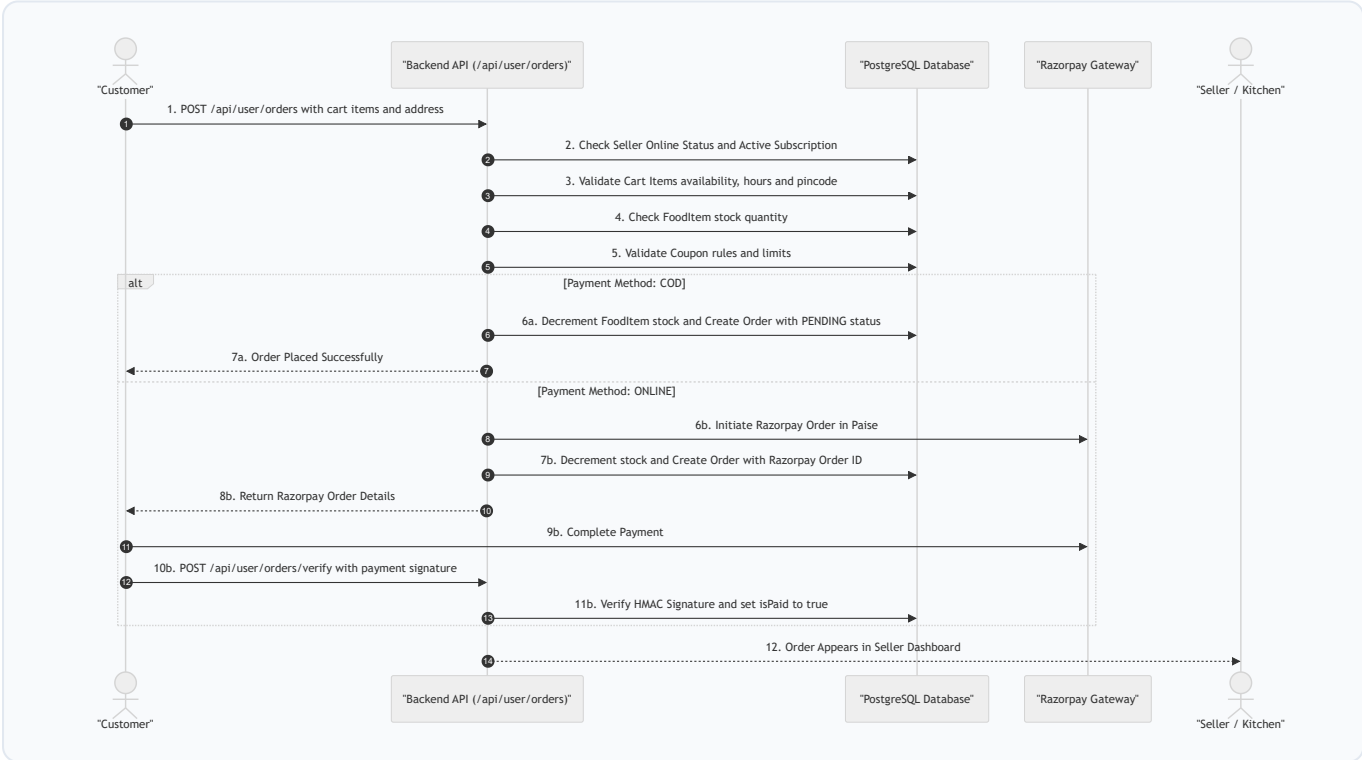
Workflow 1: Registration, Verification & Reapply / Revision Flow



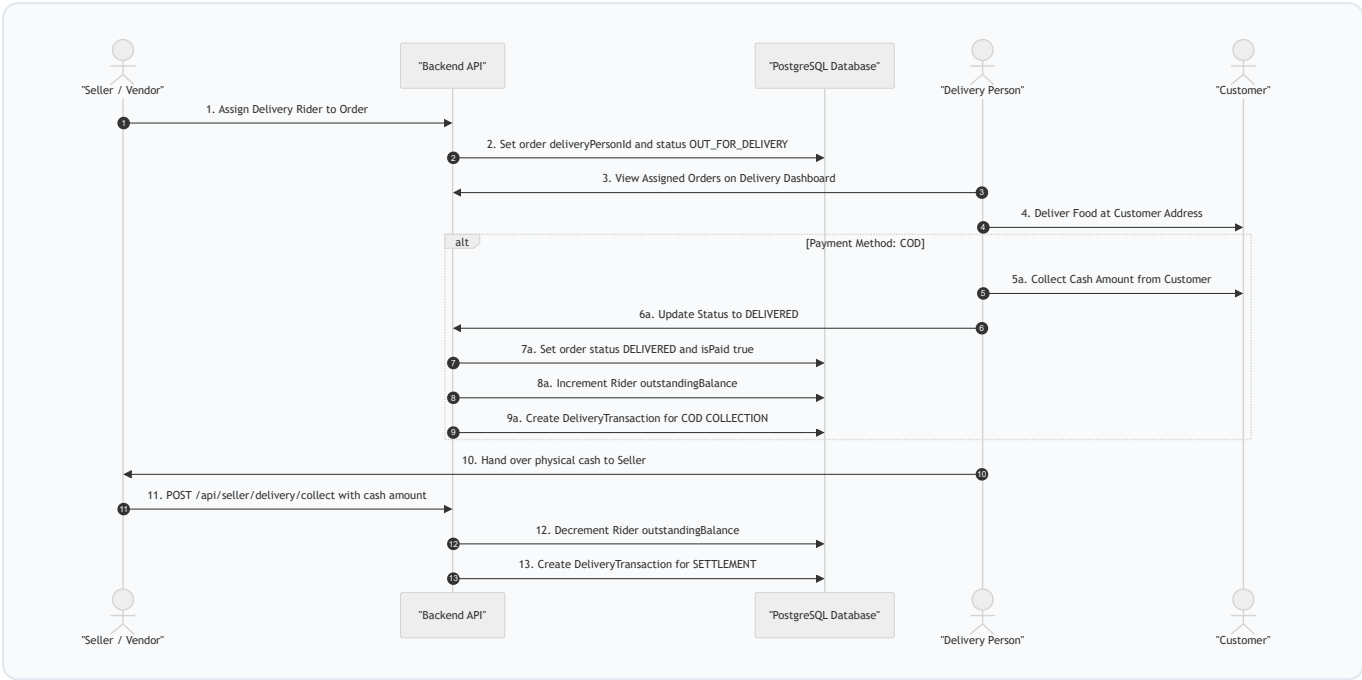
Workflow 2: Seller Subscription & Plan Renewal Stacking



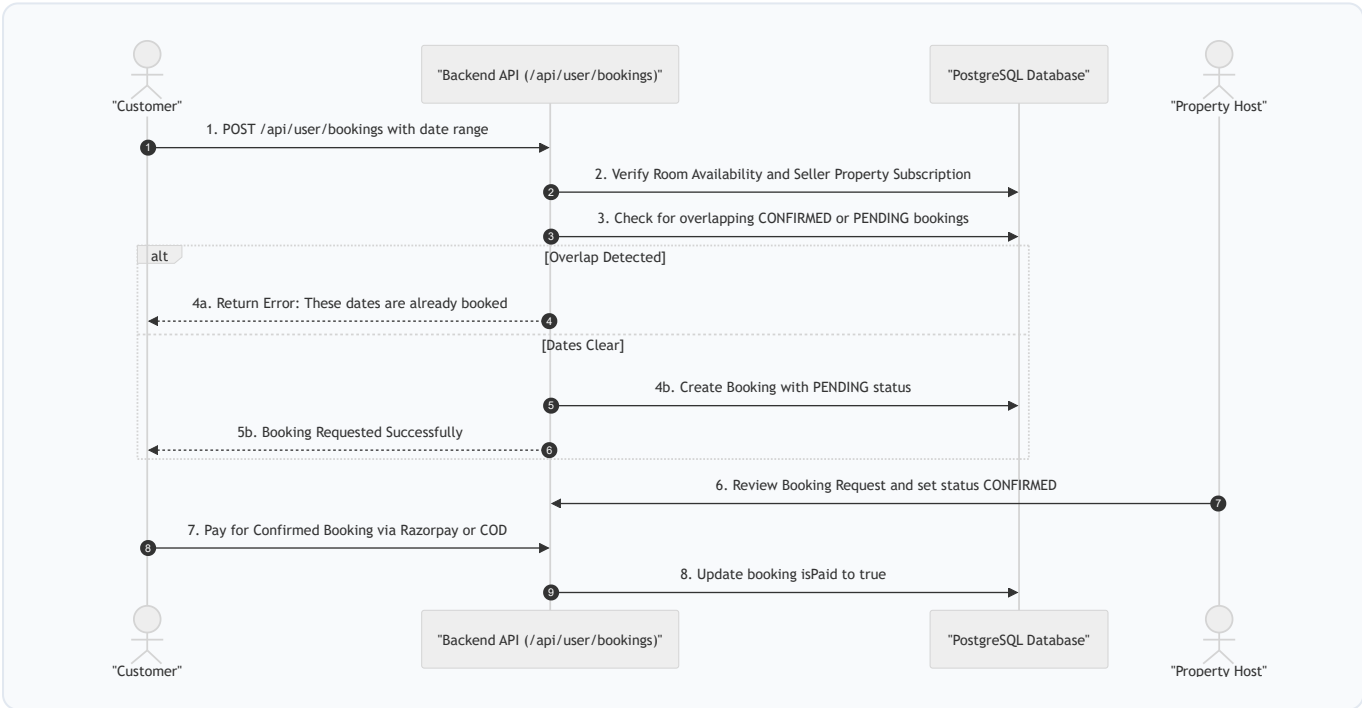
Workflow 3: Customer Food Ordering & Real-Time Stock Validation Flow



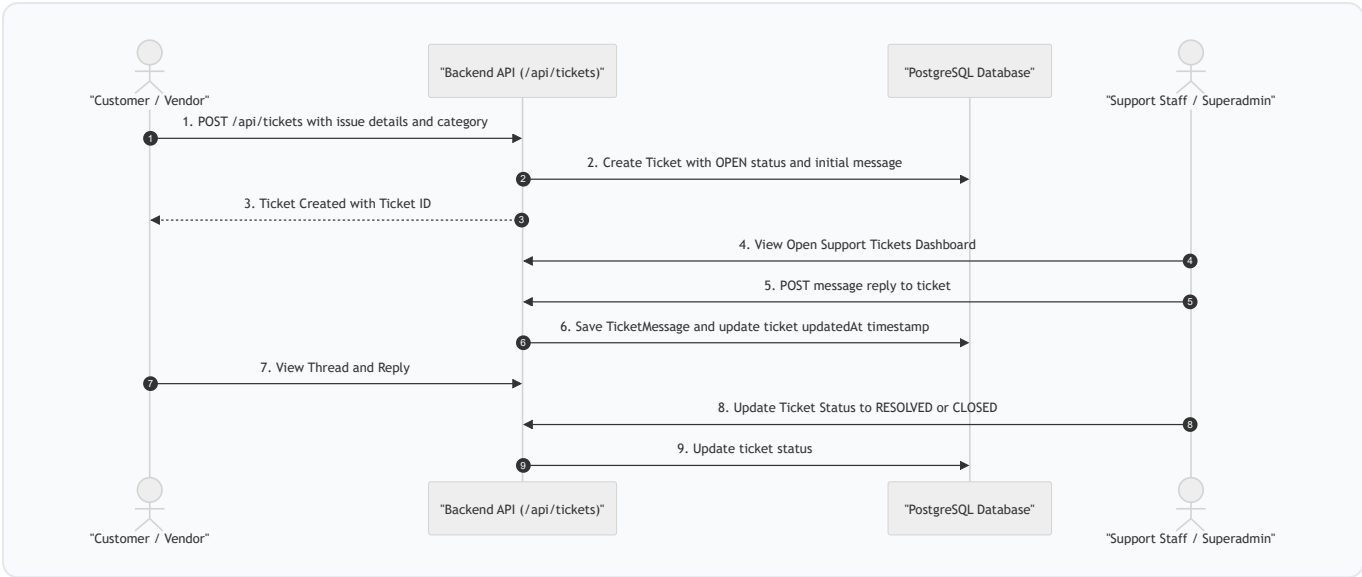
Workflow 4: Delivery Assignment, Doorstep COD Collection & Seller Cash Settlement



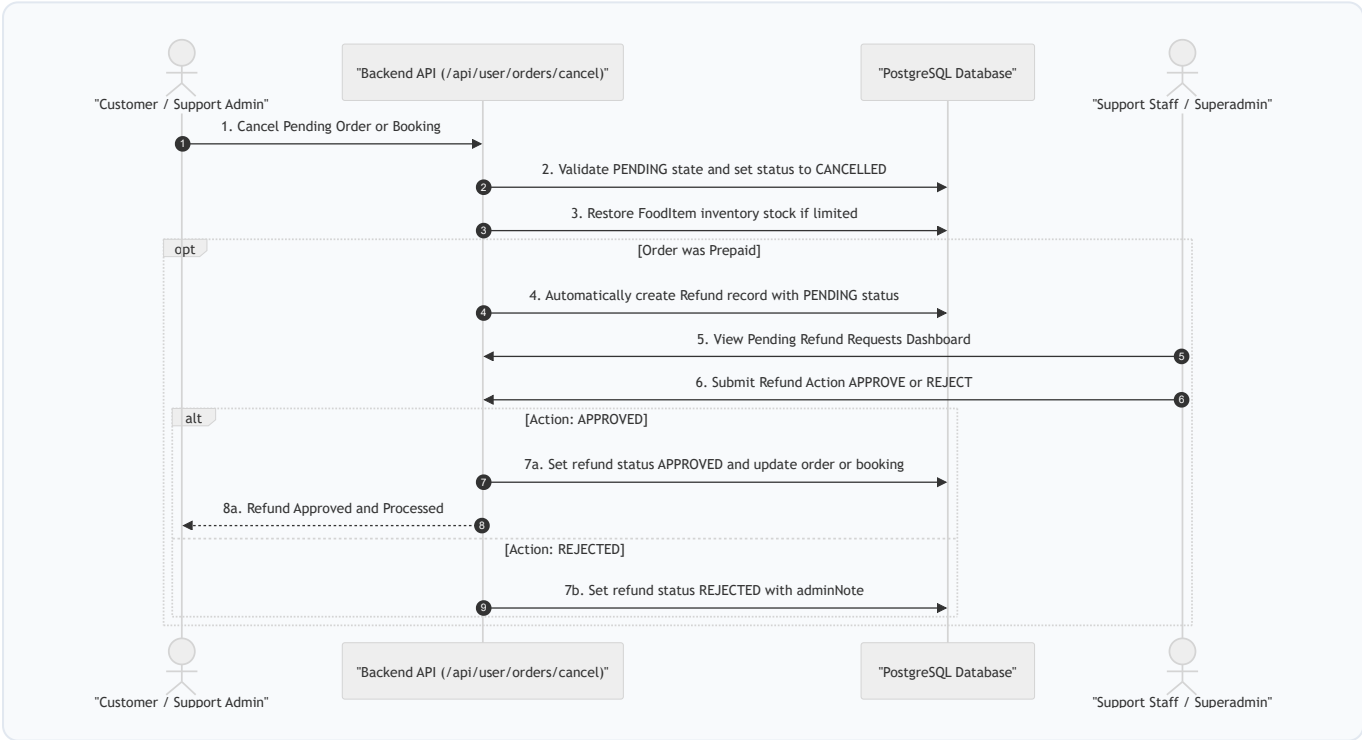
Workflow 5: Property Room Search, Booking, Date Overlap Check & Settlement Flow



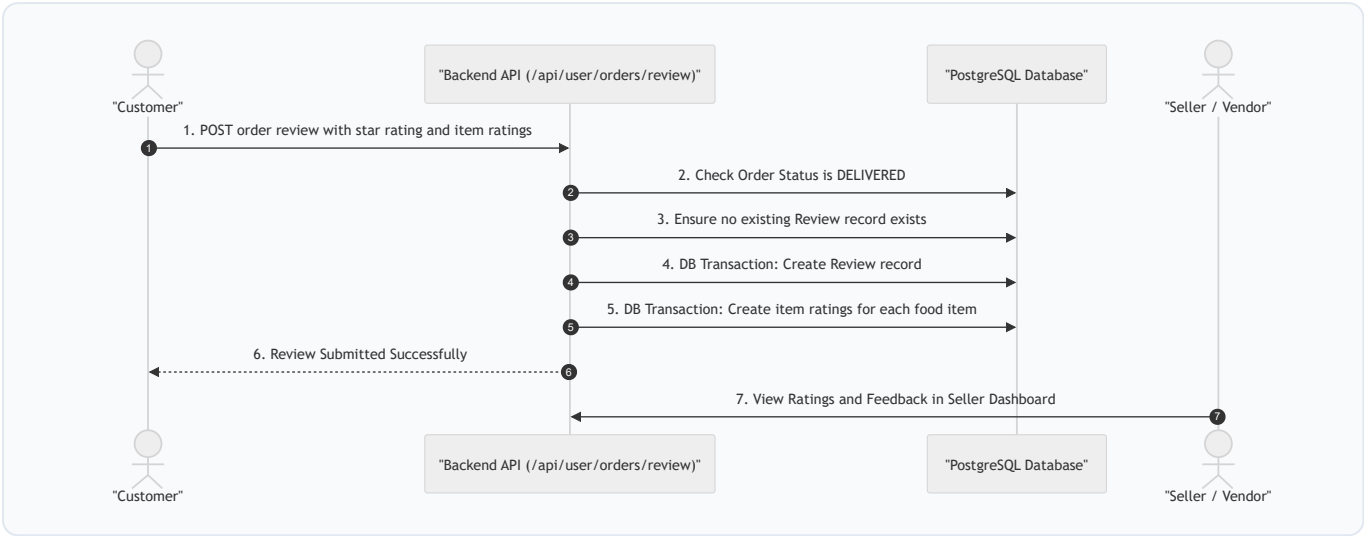
Workflow 6: Support Ticket Lifecycle & Live Threaded Chat Flow



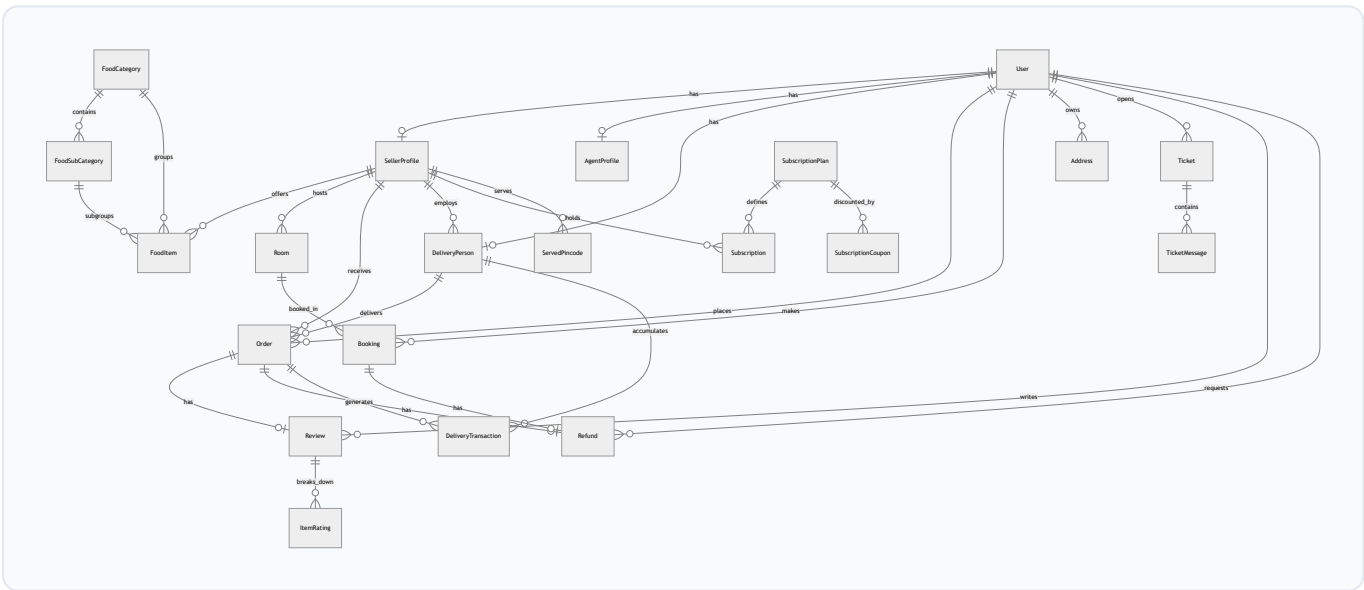
Workflow 7: Order / Booking Cancellation & Refund Approval Flow



Workflow 8: Detailed Review & Item-Level Rating Submission



5. Complete Database Models & Field Reference



Core Schema Entity Summary

- User**: Central identity model storing credentials (`passwordHash`), system role (`SUPERADMIN` , `AGENT` , `SUPPORT` , `SELLER` , `USER` , `DELIVERY`), contact details (`name` , `email` , `phone` , `city` , `pincode`), and active status (`isActive`).
- SellerProfile**: Business metadata for vendors storing type (Homely Food, Mess, Bakery), business name, flat/area/landmark address, JSON arrays of kitchen/cuisine/room photos, legal document URLs (Adhaar, FSSAI, Light Bill, Passbook), global verification state (`verificationStatus` , `verificationNote`), domain specific statuses (`foodVerificationStatus` , `propertyVerificationStatus`), tracking ID (`trackingId`), online toggle (`isOnline`), UPI ID, and assigned field Agent reference (`agentId`).
- ServedPincode**: Unique mapping of pincodes served by each seller.

4. **AgentProfile:** Field admin permissions (`canManageOffers` , `canManageBanners`) and relation to assigned sellers.
 5. **Category / FoodCategory / FoodSubCategory:** Hierarchical menu taxonomies organizing food items and property types.
 6. **FoodItem:** Individual dish schema storing name, description, price, food type (`VEG` / `NON_VEG`), image URL, available days, stock quantity (-1 for unlimited), operational hours JSON object, delivery pincodes string, and categories.
 7. **Room:** Property accommodation listing storing title, description, capacity, night price, photos array, and availability boolean (`isAvailable`).
 8. **Order:** Food purchase schema storing user ID, seller ID, assigned delivery person ID, JSON serialized items array, status (`PENDING` , `PREPARING` , `OUT_FOR_DELIVERY` , `DELIVERED` , `CANCELLED`), delivery address, customer phone, payment method (`ONLINE` , `COD` , `QR`), total amount, paid flag (`isPaid`), applied coupon ID, Razorpay order/payment IDs.
 9. **Booking:** Room rental transaction storing room ID, user ID, start/end dates, status (`PENDING` , `CONFIRMED` , `CANCELLED`), payment method, total amount, paid flag, and Razorpay references.
 10. **SubscriptionPlan & Subscription:** Vendor monetization models defining monthly price, category (`FOOD` , `PROPERTY` , `BOTH`), features list, validity start/end timestamps, Razorpay transaction links, and applied subscription coupon.
 11. **SubscriptionCoupon & Coupon:** Store and platform promotion engine storing discount percentage or flat amount, maximum usage limits, current usage counts, minimum cart value thresholds, and approval states.
 12. **Address:** Customer delivery address book storing house number, street, landmark, pincode, latitude, longitude, and default marker (`isDefault`).
 13. **PopupBanner:** Promotional banner schema storing image URL, redirect URL, targeted seller store ID (or global), and approval status (`approvalStatus`).
 14. **DeliveryPerson:** Delivery driver profile storing assigned seller ID, contact info, active status, and live COD cash balance (`outstandingBalance`).
 15. **DeliveryTransaction:** Wallet ledger tracking rider cash activity (`COD_COLLECTION` , `SETTLEMENT` , `ADJUSTMENT`) with amount and descriptions.
 16. **Review & ItemRating:** Customer feedback schemas capturing overall order star ratings (1-5), written comments, and granular item-level star ratings.
 17. **Ticket & TicketMessage:** Support ticketing engine storing ticket title, description, category, status (`OPEN` , `IN_PROGRESS` , `RESOLVED` , `CLOSED`), and threaded chat messages.
 18. **Refund:** Financial refund tracking schema for paid orders or bookings storing requested amount, cancellation reason, status (`PENDING` , `APPROVED` , `REJECTED`), and administrative notes.
 19. **AuditLog:** Security audit logging table recording administrative actions (`action` , `performedBy` , `details` , `timestamp`).
 20. **SystemSettings:** Key-value pair configuration table for platform-wide settings.
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6. Summary & System Features Checklist

Feature Category	Capabilities Included in Platform
Authentication & Security	Multi-role JWT & NextAuth authentication, Password hashing with bcrypt, HMAC-SHA256 Razorpay payment signature validation, Role-based API route protection.
Vendor Onboarding	Multi-document upload (Adhaar, FSSAI, Light Bill, Passbook), Multi-photo store galleries, Tracking ID generation, Administrative audit review with resubmission notes (REVISION state).
Vendor Reapply & Expansion	Document resubmission pipeline for rejected/revision profiles, Category expansion application (FOOD → PROPERTY & vice versa) without re-registering.
Monetization & Subscriptions	Category-based subscription plans (Food vs Property), Stacking validity renewals, Subscription coupon discounts, Automatic store lock upon subscription expiry.
Food Engine & Inventory	Stock quantity tracking with auto-decrement on order and auto-restore on cancellation, Day-wise operational hours validation (IST timezone), Pincode delivery matching.
Property & Room Bookings	Calendar date range selection, Automated date overlap validation against existing bookings, Two-stage approval (Host Confirmation → Payment).
Promotions & Offers	Store-level vs Global coupons, Minimum cart value rules, Max per-user and max global redemption limits, Store pop-up banners with Superadmin approval workflow.
Delivery & COD Wallet	Seller-managed rider fleet creation, Automated rider COD balance increment upon delivery, Cash settlement collection logging (SETTLEMENT), Administrative balance adjustments (ADJUSTMENT).
Customer Support & Refunds	Support ticket creation across categories, Live threaded chat messages, Support staff order cancellation overrides, Automated refund creation & fulfillment (APPROVED / REJECTED).
Ratings & Quality Control	Overall 1-5 star order reviews, Granular item-by-item star ratings (ItemRating), Seller review dashboard.

Documentation compiled and generated for the Neo Cloud Room / CloudKitchen codebase.